



UNIVERSITY OF THE PUNJAB

Fourth Semester 2015

Examination: B.S. 4 Years Programme

Roll No.

PAPER: Statistics-IV

Course Code: STAT-203 /

TIME ALLOWED: 2 hrs. & 30 mins.

MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Q.II Attempt following question. (20)

- Write down the properties of chi-square distribution.
- How is the chi-square independence test similar to goodness of fit? How is it different?
- Write the assumptions of multiple linear regression model.
- Differentiate between multiple and partial correlations.
- Define multiple coefficient of determination? What does it measure?
- What are the assumptions under lying analysis of variance?
- Discuss why it is not a good statistical procedure to perform several t-test pairs of means when several means are to be compared suggest the alternate procedure
- What are Basic Experimental designs?
- What are the basic principles of an experimental design.
- Define Regression and correlation.

Q.III Following is the record of the no. of male and female births in families having children. Test whether the data are consistent with hypothesis that binomial law holds with the chance of male birth as 0.57: (06)

No. of male birth	0	1	2	3	More than 3
No. of families	30	180	290	200	60

Q.IV From a sample of 200 pairs of observations the following calculations are made which follow $Y = \alpha + \beta X + \epsilon$ (06)

$$\Sigma X = 11.34, \Sigma Y = 20.72, \Sigma X^2 = 12.16, \Sigma Y^2 = 84.96, \Sigma XY = 22.13$$

Find 95% confidence limits for mean value of Y when $X_c = 0.10$

Q.V Following are the treatment and block totals and other calculations in a RCB design. (06)

Treatments totals: 286, 293, 240, 404, 322

Block totals: 361, 428, 407, 349

Total sum of squares uncorrected is 124069. Perform analysis and draw conclusions.

Q.VI The following calculations were obtained from 10 sets of observations which follow $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$ (06)

$$\Sigma Y = 20, \Sigma X_1 = 30, \Sigma X_2 = 40, \Sigma Y^2 = 88.2, \Sigma X_1^2 = 92,$$

$$\Sigma X_2^2 = 163, \Sigma X_1 Y = 59, \Sigma X_2 Y = 88, \Sigma X_1 X_2 = 119.$$

Find co-efficient of determination of the above model.

Q.VII Following are the calculations in a L.S. Design. Row SS = 46.67, (06)
Column SS = 14.02, Treatment SS = 196.61, Total SS = 285.34, Treatment means are 6.84, 6.46, 13.12, 7.96, 4.92. Perform analysis and apply lsd test.

$$\frac{1}{(2)^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)}$$

question 2(1)

Properties of Chi-Square Distribution :

1. The chi-square is a continuous distribution ranging from zero to plus infinity i.e. $0 < \chi^2 < \infty$.
2. The curve of a χ^2 -distribution is positively skewed.
3. The total area under the curve of χ^2 -distribution is unity.
4. The mean of chi-square distribution is equal to the number of degree of freedom i.e. $\mu = E(\chi^2) = n$ and variance is equal to twice the number of degree of freedom, i.e. $\text{Var}(\chi^2) = 2n$.
5. The third and fourth moments about mean of Chi-square distribution are

question 2(1)

$$\mu_3 = 8n, \mu_4 = 12n^2 + 48n, \beta_1 = \frac{8}{n}, \beta_2 = 3 + \frac{12}{n}$$

6. The χ^2 -distribution tends to normal distribution as the number of degree of freedom approaches infinity.
7. Chi-square distribution has one parameter, the degree of freedom.

The Chi-square test for independence looks for an association between two categorical variables within the same population. Unlike the goodness of fit test, the test for independence does not compare a single observed variable to a theoretical population, but rather two variables within a sample set to one another.

question 2(2)

question 2(3)

iii)

- 1) All linear relationship between the dependent and independent variables
- 2- The independent variables are not highly correlated with each other.
- 3- The variance of the residuals is constant.
- 4- Independence of observation
- 5- Multivariate normality

Multiple correlation

- It is the simple correlation b/w variable and as estimate of that variable obtained from a linear equation involving two or more independent variables.

- Denoted by R .

Partial correlation

- It is the simple correlation b/w variable and one independent variable after adjusting each for the effect of one or more other variables.

- Denoted by r_{xy} .

question 2(4)

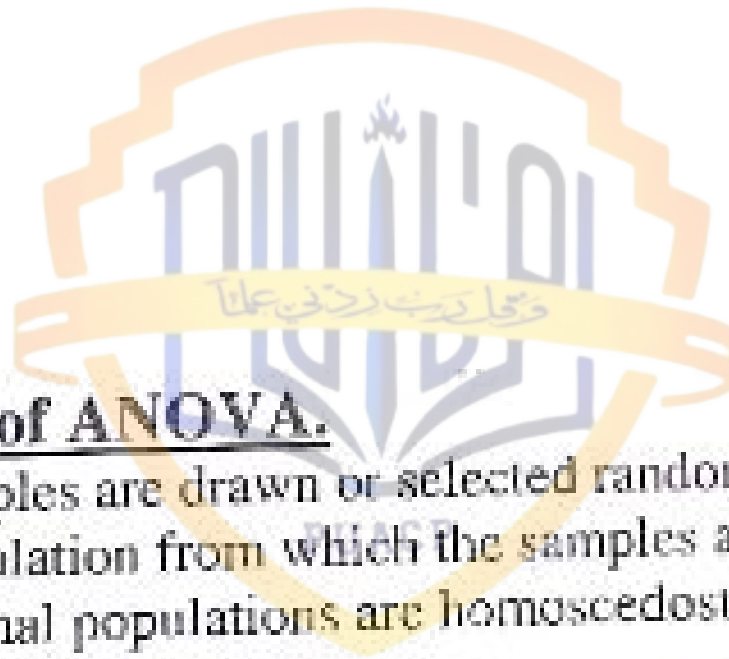
Coefficient of Multiple Determination

question 2(5)

The coefficient of multiple determination, denoted R^2 , in multiple regression is similar to the coefficient of determination in simple linear regression, except in multiple regression there is more than one independent variable. The coefficient of multiple determination is the proportion of variation in the dependent variable that can be explained by the multiple regression model based on the independent variables.

question 2(5)

It tells us how differences in one variable can be explained by the difference in a second variable, while predicting the outcome of given event.



Assumptions of ANOVA.

1. The samples are drawn or selected randomly and independently from the population.
2. The population from which the samples are drawn are normally distributed.
3. The normal populations are homoscedosticity i.e. (have equal variances).
4. The effects are additive.

question 2(6)

~~They are equivalent.~~ An ANOVA with only two groups is equivalent to a t-test. The difference is when you have several groups then the type I error will increase for the t-tests as you are not able to test the hypothesis jointly. ANOVA does not suffer from this problem as you jointly test them through an F-test. **question 2(7)**



Basic Experimental Designs.

There are three basic experimental designs.

1. Completely Randomized Design (CRD)
2. Randomized Complete Block Design (RCBD)
3. Latin Square Design (LS Design)

question 2(8)

Basic Principles of Experimental Designs.

There are three basic principles of experimental designs.

1. Randomization, 2. Replication, 3. Local Control

1. Randomization.

A random process of allocation of treatments to the experimental units is known as randomization. It is used to reduce the experimental error.

2. Replication.

The repetition of an experiment is called replication. It reduces the experimental error.

3. Local Control or Error Control.

The term error control or local control refers to the amount of balancing, blocking and grouping of the experimental units. Its main object is to increase the efficiency of an experimental design.

question 2(9)

question 2(10)

Q7.2 Write short answers to the following questions in the space provided underneath.

Q.1. Differentiate between regression analysis and correlation analysis.

(Federal Board Islamabad 2010)

Ans. In the simple regression analysis we obtain a regression equation which can be used to estimate the values of the dependent variable on the basis of the independent variable whose values are known. In correlation analysis we study the degree of closeness of relationship between the variables. In case of regression the dependent variable is assumed to be *fixed* or *known*; in case of correlation both the variables are assumed to be *random*. Correlation simply means the co-variation of variables. It does not mean that changes in one variable cause a change in the other.

question 3

① Hypothesis:-

H_0 : The given distribution is Normal

H_1 : The given distribution is not Normal

$$\alpha = 0.05$$

$$\frac{\sum X}{f} = \frac{1600}{760}$$

② $1 - 0.5$:-

③ Test statistic:-

$$\chi^2 = \frac{\sum (O_i - E_i)^2}{E_i}$$

$$p = 0.52$$
$$q = 0.43 \quad 216$$

④ Calculation:-

X	f	$P(X=x)$	$E = N P(X)$	$(O_i - E_i)^2 / E_i$
0	30	0.01948	14.80	15.61
1	180	0.18129	137.76	12.951
2	290	0.3604	273.90	9.916
3	200	0.3185	242.06	7.308
4	60	0.1055	80.18	5.078
10	760			

5) Critical region: We will reject H_0 if

$$\chi^2_c \geq \chi^2_{(k-1-m)}$$

$$\chi^2_c \geq \chi^2_{0.05(5-1-0)}$$

$$50.855 \geq 9.49$$

question 3

6) Conclusion:-

As Calculated value of χ^2 lies in critical region
So we will not reject H_0